GOWTHAM M

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PROFESSIONAL SUMMARY

Results-oriented AI/ML student pursuing a Bachelor of Technology in Artificial Intelligence and Data Science, with strong proficiency in Python and machine learning fundamentals. Experienced in building IoT-integrated machine learning applications, multi-agent LLM systems, and ETL data pipelines, including object detection projects. Seeking to apply strong analytical skills and a collaborative mindset to deliver efficient, production-ready software solutions.

SKILLS

Technical Skills: Python, SQL, Git/GitHub, Machine Learning, Deep Learning, IoT Integration, Data Analysis, LangChain, LangGraph, Generative AI / LLM Application Development

Tools & Concepts: PyTorch, Apache Airflow, Cloud Deployment (familiarity), Data Pipelines, Object Detection, Recommendation Systems, Ollama (local LLM deployment), Prompt Engineering, Multi-Agent System Design

Soft Skills: Teamwork, Communication, Problem Solving, Time Management

EDUCATION

Bachelor of Technology - Artificial Intelligence and Data Science

Mahendra Engineering College | CGPA: 7.45

Golden Spark Matriculation Higher Secondary School

HSC: 75% | SSLC: 77%

ACADEMIC PROJECTS

Multi-Agent Resume / Job-Matching System — LangGraph, LangChain, Python

- Built a 7-agent LangGraph pipeline covering resume parsing, ATS scoring, and job-fit recommendations.
- Used LangChain with Pydantic for typed, structured data exchange between agents.
- Deployed the system fully locally with Ollama, avoiding paid API dependency.

ETL Pipeline with Apache Airflow — Python, MySQL, SQL

- Automated a MySQL to Python to CSV pipeline using Apache Airflow.
- Scheduled the pipeline to run every 5 minutes with built-in error handling.
- Validated extracted data and managed database connections for reliability.

AI-Based Smart Traffic Signal Control Using Helmet Detection and IoT

- Built a deep learning model to detect helmet usage in real time.
- Linked detection results to IoT-based traffic signal control logic.
- Cleared signals only for rule-compliant riders, improving road safety.

IoT-Based Smart Car Parking System — Machine Learning, IoT

- Built a machine learning model to detect available parking spaces.
- Used IoT sensors to guide vehicles to open spots.
- Reduced search time and improved space utilization.

Housing ETL Project — Python, SQL

- Built a pipeline to extract, clean, and transform housing market data.
- Loaded processed data into a structured format for analysis.
- Automated data cleaning and validation using SQL and Python.

Window Info Collector App — Python

- Built an application to capture and log active window information.
- Tracked application activity for system usage monitoring.
- Organized logged data for easy retrieval and review.